

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – PSEUDOCODE WRITING TASK

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO5 – Naturalization (BL4, BL6)	Assessment:	Assessment Tool 1 – Pseudocode Writing Task
Weightage:	35% of CIA	Total Marks:	100
CO5 Statement:	<i>Construct MST using shortest path algorithms and traverse the graph to model and analyse realworld problem.</i>		
Student Name:	Sankar R	Reg. No.:	192525131
Section / Batch:	Slot B	Date:	28/08/2026

Code of Conduct

I Sankar R(192525131) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: sankar R_____

Instructions

- This test contains 5 Pseudocode Writing Task questions. Total: 100 Marks.
- When a student answer using pseudocode writing task should evaluate based on the ability to design clear, logical, and efficient pseudocode solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20

Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and clear understanding; handles edge cases effectively	Good understanding; minor gaps in edge case handling	Partial understanding; misses some requirements	Misinterprets problem; major gaps
Code Correctness	30	Fully correct; passes all test cases	Mostly correct; minor errors	Partially correct; several test cases fail	Incorrect or nonfunctional code
Code Quality & Readability	20	Clean, modular, wellcommented, follows standards	Readable with some structure	Limited readability; poor structure	Unorganized and hard to read
Complexity Analysis	20	Accurate time & space complexity with justification	Correct but limited explanation	Incomplete or partially correct	Missing or incorrect
Testing & Validation	10	Thorough testing including edge cases	Adequate testing with few edge cases	Minimal testing	No testing evidence
Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	

SIMATS
ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

SANKAR R

192525131RB

PSEUDOCODE WRITING TASK

← Back

QUESTIONS

Q1

Minimum Spanning Tree -
Kruskal's Algorithm

20 marks

Q2

Graph - Topological Sort

20 marks

Q3

Shortest Path Algorithm - MST
& Dijkstra's Algorithm

20 marks

Q4

Shortest Path Algorithm -
Dijkstra's Algorithm

20 marks

Q5

Minimum Spanning Tree -
Prim's or Kruskal's Algorithm

20 marks

5/5 answered

Q1 Minimum Spanning Tree - Kruskal's Algorithm

20 marks

Write pseudocode for constructing an MST using Kruskal's Algorithm. A city planning department needs to connect multiple zones with underground fiber-optic cables. Each zone is represented as a vertex, and possible cable connections between zones are represented as edges with installation costs (weights).

Constraints:

- All zones must be connected
- Total installation cost must be minimized
- No redundant connections (no cycles) allowed
- The network must remain fully connected

Your Answer

5. Repeat step 4 until MST_Edges contains (V - 1) edges
(this ensures all zones are connected without redundant connections)

6. Return MST_Edges as the final Minimum Spanning Tree

End Algorithm

Save Answer

Next →

SIMATS Online Lab

Ask Gemini

← → ↺

Not Secure

simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=517

🔍 ☆ 📄 🔄 👤 ⋮

SIMATS
ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

SANKAR R
192525131RB

PSEUDOCODE WRITING TASK

← Back

QUESTIONS

Q1

Minimum Spanning Tree -
Kruskal's Algorithm

20 marks

Q2

Graph - Topological Sort

20 marks

Q3

Shortest Path Algorithm - MST
& Dijkstra's Algorithm

20 marks

Q4

Shortest Path Algorithm -
Dijkstra's Algorithm

20 marks

Q5

Minimum Spanning Tree -
Prim's or Kruskal's Algorithm

20 marks

5/5 answered

Q2 Graph - Topological Sort

20 marks

A DAG represents task dependencies. Write pseudocode for Topological Sort using DFS with cycle detection.

Constraints:

a. Detect cycle before sorting

b. Use DFS approach

Your Answer

TOPOLOGICAL_SORT(Graph G):
 // Initialize data structures
 visited = array of size V, all set to WHITE (unvisited)
 stack = empty stack
 hasCycle = false

Save Answer

Next →

SIMATS
ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

SANKAR R

192525131RB

PSEUDOCODE WRITING TASK

← Back

QUESTIONS

Q1

Minimum Spanning Tree -
Kruskal's Algorithm

20 marks

Q2

Graph - Topological Sort

20 marks

Q3

Shortest Path Algorithm - MST
& Dijkstra's Algorithm

20 marks

Q4

Shortest Path Algorithm -
Dijkstra's Algorithm

20 marks

Q5

Minimum Spanning Tree -
Prim's or Kruskal's Algorithm

20 marks

5/5 answered

Q3 Shortest Path Algorithm - MST & Dijkstra's Algorithm

20 marks

Write pseudocode integrating MST + Dijkstra for Smart transportation system.

Constraints:

a. Use MST for infrastructure

b. Use shortest path for routing

Your Answer

ALGORITHM SmartTransportationSystem

// Step 1: Infrastructure Setup using Minimum Spanning Tree (MST)

// Goal: Connect all locations with minimum total road length/cost

FUNCTION BuildInfrastructure(Graph G):

Initialize empty set MST_Edges

Save Answer

Next →

SIMATS

ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

SANKAR R

192525131RB

Not Secure

simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=517

Ask Gemini

PSEUDOCODE WRITING TASK

← Back

QUESTIONS

Q1

Minimum Spanning Tree -
Kruskal's Algorithm

20 marks

Q2

Graph - Topological Sort

20 marks

Q3

Shortest Path Algorithm - MST
& Dijkstra's Algorithm

20 marks

Q4

Shortest Path Algorithm -
Dijkstra's Algorithm

20 marks

Q5

Minimum Spanning Tree -
Prim's or Kruskal's Algorithm

20 marks

5/5 answered

Q4 Shortest Path Algorithm - Dijkstra's Algorithm

20 marks

Shortest Path with "Toll-Free" Voucher: You have a voucher that allows you to traverse one edge for free (weight = 0). Write pseudocode to find the shortest path from S to D utilizing this voucher optimally.

Your Answer

Push (dist1[v], v, 1) to PQ

ii. If state == 1 (Voucher already used):

* If $\text{dist1}[u] + w < \text{dist1}[v]$:

$\text{dist1}[v] = \text{dist1}[u] + w$

Push (dist1[v], v, 1) to PQ

5. Return $\min(\text{dist0}[D], \text{dist1}[D])$

Save Answer

Next →

